TECHNICAL PAPER REVIEW FORM

I. COMMENTS TO AUTHORS (WITHIN 300 WORDS)

THIS PAPER PROPOSES A COORDINATED EMERGENCY-RESPONSE FRAMEWORK FOR HAZARDOUS-MATERIAL VEHICLE INCIDENTS ON URBAN EXPRESSWAYS, INTEGRATING EMERGENCY SUPPLY DISTRIBUTION AND MULTI-TYPE PERSONNEL EVACUATION UNDER A CHLORINE-RELEASE SCENARIO ON BEIJING'S SIXTH RING ROAD. USING GAUSSIAN PLUME MODELLING, ERPG-BASED ZONING, A TRUCK-DRONE LOGISTICS NETWORK, AND MULTIMODAL EVACUATION DISPATCH, THE PAPER SHOWS THAT DIFFERENTIATED ZONING CAN SUBSTANTIALLY REDUCE THE ACTIVE EVACUATION POPULATION, THAT VULNERABLE GROUPS CAN BE EFFECTIVELY PROTECTED, AND THAT MEDICAL EVACUATION CAPACITY IS THE MAIN SYSTEM BOTTLENECK. THE PAPER IS RELEVANT TO THE CONFERENCE SCOPE AND ADDRESSES AN IMPORTANT PROBLEM IN EMERGENCY TRANSPORTATION MANAGEMENT. ITS MAIN STRENGTHS ARE THE INTEGRATED PERSPECTIVE, THE REALISTIC URBAN CASE SETTING, AND THE CONNECTION BETWEEN OPTIMIZATION RESULTS AND RESILIENCE-ORIENTED POLICY IMPLICATIONS.

HOWEVER, SEVERAL REVISIONS ARE NEEDED. THE MOST IMPORTANT CONCERN IS RESULT CONSISTENCY AND METHODOLOGICAL TRANSPARENCY. (1)IN SECTION 4.3, THE TEXT REPORTS THAT THE TRUCK-ONLY SCHEME COMPLETES 64 TASKS AND THE TRUCK-DRONE SCHEME COMPLETES 62 TASKS, WHILE TABLE 2 REPORTS 61 AND 60, RESPECTIVELY. (2)THE OPTIMIZATION PROCEDURES ARE DESCRIBED ONLY AT A HIGH LEVEL. THE PAPER SHOULD MORE CLEARLY EXPLAIN THE OBJECTIVE-FUNCTION COMPONENTS, WEIGHT SETTINGS, RELAY-TRIGGER RULE, UNMET-DEMAND PENALTY, AND HOW THE GREEDY + 2-OPT ALGORITHM IS IMPLEMENTED. SEVERAL KEY PARAMETERS ALSO REQUIRE JUSTIFICATION, INCLUDING THE 5 KG/MIN CHLORINE RELEASE RATE, THE ASSUMED 2% K=1 CRITICALLY INJURED SHARE IN THE ERPG-1 ZONE, FLEET SIZES, AND INVENTORY CAPACITIES. (3) THE RESILIENCE DISCUSSION IS INSIGHTFUL, BUT THE PAPER CURRENTLY USES "RESILIENCE" MAINLY AS AN INTERPRETIVE CONCEPT; THE INDICATORS AND EVALUATION LOGIC SHOULD BE STATED MORE EXPLICITLY. (4)BECAUSE CONCLUSIONS ARE BASED ON ONE HYPOTHETICAL SCENARIO, THE PAPER WOULD BENEFIT FROM LIMITED ADDITIONAL SENSITIVITY ANALYSIS OR STRONGER DISCUSSION OF SCENARIO DEPENDENCE.

II. RECOMMENDATIONS

OVERALL RECOMMENDATION

☒ ACCEPT

☐ ACCEPT AFTER MINOR REVISION (WITHOUT RE-REVIEW)

☐ TO BE REVISED AND RESUBMITTED (MAJOR REVISION)

☐ REJECT

PRESENTATION RECOMMENDATION

☒ ORAL PRESENTATION

☐ POSTER PRESENTATION